Normality of squarefree-weighted series

Stefan Ringer*

27 September 2026

Erdős problem 259 [2] asks whether

$$\sum_{n\geq 1}\frac{n\mu(n)^2}{2^n}$$

is irrational, where μ is the Möbius function, so that $\mu(n)^2$ is the indicator of squarefree integers. Chen and Ruzsa [1] proved this, including a stronger statement for infinite subseries. We establish normality of the full series and extend the result to every integer denominator base.

A real number is normal to base B if each word of ℓ digits occurs with limiting frequency $B^{-\ell}$. For $x_1,\dots,x_M\in[0,1)$, write

$$D_M(x_1,\dots,x_M)=\sup_{0\leq a<b\leq 1}\left|\frac{1}{M}\#\{1\leq n\leq M : x_n\in[a,b)\}-(b-a)\right|.$$

Here $\{x\}$ denotes the fractional part of x. We use $\mathrm{e}(t)=\exp(2\pi it)$, $\|t\|=\operatorname{dist}(t,\mathbb{Z})$, and $[t]_+=\max(t,0)$. All Fourier frequencies are integers, and logarithms are natural unless their base is indicated.

Theorem 1. *For every fixed integer $B\geq 2$, the number*

$$\alpha_B=\sum_{n\geq 1}n\mu(n)^2B^{-n}$$

is normal to base B, with

$$D_M((\{B^n\alpha_B\})_{n\leq M})\ll_B(\log\log M)^{-1}.$$

The base B and the number α_B vary together; no assertion about absolute normality of a fixed α_B is made. The proof approximates squarefreeness by finitely many divisibility conditions. On each resulting residue class, the orbit phase is affine in its index. An average over integer frequencies controls the classes whose slopes are close to integers.

Write $\vartheta(t)=\sum_{p\leq t}\log p$ and $\pi(t)=\#\{p\leq t : p \text{ prime}\}$. We use the standard consequences of the prime number theorem

$$\vartheta(y)\sim y,\qquad \sum_{p>z}p^{-2}\ll(z\log z)^{-1},\qquad \pi(2z)-\pi(z)\gg z/\log z. \tag{1}$$

*AI tools (GPT 6 Astra and Fable 5.1) assisted with the development and checking of the proofs.

A Fourier estimate for the periodic sieve. For large y put

$$Q=\prod_{p\le y}p^2,\quad s_y(n)=\prod_{p\le y}\mathbf{1}_{p^2\nmid n},\quad A_r=\sum_{j\ge 1}s_y(r+j)B^{-j},\quad \widehat{\nu}_y(m)=\frac{1}{Q}\sum_{r \bmod Q}\mathrm{e}(mA_r).$$

For an integer $L\ge 1$ and a consecutive interval $W\subseteq\{1,\dots,L\}$ set

$$E_W(m)=\sum_{j\in W}(1-\cos(2\pi mB^{-j})),\quad Z_W=\sum_{p\le\min(\sqrt{2L},y)}\lceil |W|/p^2\rceil,\quad V=\sum_{\sqrt{2L}<p\le y}p^{-2}.$$

Lemma 1. *For every nonzero integer m,*

$$|\widehat{\nu}_y(m)|\le e^{-\frac{1}{2}V[E_W(m)-2Z_W]_+}+LV^2+\frac{2\pi|m|}{B-1}B^{-L}.$$

Proof. Take r uniformly modulo Q and truncate A_r at L. The Chinese remainder theorem makes the residues $r \bmod p^2$ independent and uniform. Condition on those for $p\le\min(\sqrt{2L},y)$, and let u_j indicate that none of these prime squares divides $r+j$. At most Z_W entries in W vanish. For each remaining prime, at most one of the L congruences $r+j\equiv 0 \pmod{p^2}$ holds. Each has probability p^{-2}; their complement has probability $1-L/p^2>1/2$.

The probability that any position satisfies two such congruences is at most $LV^2/2$. In the products over these remaining primes, replacing $\prod_p(1-\mathbf{1}_{p^2\mid r+j})$ by $1-\sum_p\mathbf{1}_{p^2\mid r+j}$ is exact outside this event. The two resulting complex exponentials have modulus one, so their averages differ by at most LV^2. The latter average factors, up to a constant of modulus one, as a product of

$$w_p=1-L/p^2+p^{-2}\sum_{j\le L}\mathrm{e}(-mu_jB^{-j}).$$

For $\mathcal{E}_u=\sum_{j\le L}u_j(1-\cos(2\pi mB^{-j}))$, retaining the pairs involving the complementary outcome in the nonnegative pair expansion of $1-|w_p|^2$ gives $1-|w_p|^2\ge 2(1-L/p^2)\mathcal{E}_u/p^2\ge\mathcal{E}_u/p^2$. Thus $|w_p|\le e^{-\mathcal{E}_u/(2p^2)}$. Since each summand in the energy lies between zero and two, $\mathcal{E}_u\ge[E_W(m)-2Z_W]_+$. Multiply these bounds, average over the conditioned residues, and add the truncation and replacement errors. □

Exact averaging in an arbitrary integer base. For large y set

$$k=\lfloor y\rfloor,\quad J=B^k,\quad T=\lfloor(\log y)^2\rfloor,\quad s=\lceil\log_2 T\rceil+2,$$
$$L=k+\lceil\log_B(TQ)\rceil+\lceil 3\log_B(y+2)\rceil,\qquad W=\{s+1,\dots,k\},\quad w=k-s.$$

Here $s=o(k)$ and W is nonempty eventually. More precisely,

$$\frac{w}{L}\longrightarrow\frac{\log B}{\log B+2}>0,\qquad w\asymp_B L\asymp_B y.$$

Lemma 2. *For every fixed $B\ge 2$, uniformly over $1\le|\xi|\le T$,*

$$\frac{1}{J}\sum_{h=1}^{J}|\widehat{\nu}_y(h\xi Q)|\ll_B(\log y)^{-2}.$$

Proof. Put $A=|\xi|Q$ and $g=\gcd(A,B^k)$. For sufficiently large y, all primes dividing B are at most y, so $v_p(A)=v_p(|\xi|)+2\le\log_2 T+2\le s$ for each such prime. Consequently $g\mid B^s$. As h runs from 1 to B^k, $hA \bmod B^k$ is uniform over the multiples of g, each with multiplicity g. Conditional on a residue $r \bmod B^s$ with $0\le r<B^s$ and $g\mid r$, it is therefore

$$r+B^sU,\qquad U\text{ uniform on }\{0,\dots,B^{k-s}-1\}.$$

The w higher digits are independent and uniform, even when B is composite and $(A, B) > 1$.

For every $s < j \le k$, the conditional mean of $\mathrm{e}(hA/B^j)$ is zero, as U traverses complete periods of a nontrivial character. Thus $\mathbb{E}(E_W(hA) \mid r) = w$. Changing a digit at position $i \ge s$ changes the energy by at most

$$2\pi(B-1)\sum_{j>i} B^{i-j} \le 2\pi.$$

The Azuma–Hoeffding inequality in its bounded-differences form gives, for $a = 19/20$,

$$\mathbb{P}(E_W(hA) < aw \mid r) \le \exp\left(-\frac{(1-a)^2 w}{2\pi^2}\right) = e^{-w/(800\pi^2)}.$$

This inequality follows by exposing the independent digits one at a time: the martingale increment has conditional range at most 2π; Hoeffding's exponential estimate and Markov's inequality give the display. Average over r to obtain the same bound without conditioning.

The elementary numerical bound $\sigma = \sum_p p^{-2} \le \sum_{p\le 53} p^{-2} + 1/53 < 0.47$ implies $Z_W \le \sigma w + \pi(\sqrt{2L})$. Since $a - 2\sigma > 1/100$, outside the exponentially small exceptional set,

$$E_W(hA) - 2Z_W \ge (a-2\sigma)w - 2\pi(\sqrt{2L}) \ge \delta_B L.$$

By (1), using $(\sqrt{2L}, 2\sqrt{2L}] \subseteq (\sqrt{2L}, y]$ eventually, $V \asymp_B 1/(\sqrt{L}\log L)$. Also $hAB^{-L} \le (y+2)^{-3}$. Lemma 1 bounds the terms outside the exceptional set by $e^{-\delta'_B\sqrt{y}/\log y} + O_B((\log y)^{-2})$. On the exceptional set use $|\widehat{\nu}_y| \le 1$. Its contribution is exponentially small in w. Conjugation handles negative ξ. □

From Fourier averages to discrepancy. The nonnegative Fejér kernel

$$F_J(t) = 1 + 2\sum_{h=1}^{J}(1 - h/(J+1))\cos(2\pi h t) = \frac{1}{J+1}\left(\frac{\sin(\pi(J+1)t)}{\sin(\pi t)}\right)^2$$

is interpreted by continuity at integer t. It is at least $4(J+1)/\pi^2$ on $\|t\| \le 1/(2(J+1))$. Thus Lemma 2 gives

$$\frac{1}{Q}\#\{r : \|\xi Q A_r\| \le \eta\} \ll_B (\log y)^{-2} \quad \left(0 < \eta \le \frac{1}{2(J+1)},\ 1 \le |\xi| \le T\right). \tag{2}$$

Indeed, average the kernel over r, bound its nonconstant coefficients by their absolute values, and use the Fourier average above.

Proof of Theorem 1. For integer X let N be uniform on $(X, 2X]$, and put

$$c_B = \frac{1}{10(1+\log B)}, \quad y = c_B \log X, \quad H = \lceil 4\log_B X\rceil, \quad \eta = \sqrt{Q/X}, \quad Y = \log\log X.$$

Define

$$D(N) = \sum_{j\ge 1}(N+j)\mu(N+j)^2 B^{-j}, \qquad D_y(N) = \sum_{j\ge 1}(N+j)s_y(N+j)B^{-j}.$$

Exactly $B^N\alpha_B - D(N) \in \mathbb{Z}$. An indicator difference within the first H positions requires $p^2 \mid N+j$ for some $p > y$, $j \le H$. Since $N + j < 3X$, the exceptional fraction is at most

$$H \sum_{y < p \le \sqrt{3X}} (p^{-2} + X^{-1}) \ll_B Y^{-1}.$$

Outside this event, $|D(N) - D_y(N)|$ is at most

$$\tau_X = B^{-H}\left(\frac{2X+H}{B-1} + \frac{B}{(B-1)^2}\right) \ll_B X^{-3}.$$

For $r = N \bmod Q$ the periodic approximation is $D_y(N) = NA_r + C_r$, with $C_r = \sum_{j\geq 1} j s_y(r+j)B^{-j}$. Geometric summation in each progression $N = r + Qv$ gives

$$|\mathbb{E}\mathrm{e}(\xi D_y(N))| \ll \frac{1}{Q}\#\{r : \|\xi Q A_r\| \leq \eta\} + \frac{Q}{X\eta} + \frac{Q}{X}.$$

All parameters are compatible: (1) gives

$$Q = X^{2c_B+o(1)}, \quad J = X^{c_B \log B + o(1)}, \quad J\eta = X^{c_B(1+\log B)-1/2+o(1)} = X^{-2/5+o(1)} \longrightarrow 0.$$

Also $Q/(X\eta) = \sqrt{Q/X} \to 0$ with a fixed power saving. Therefore (2) applies, uniformly for $1 \leq |\xi| \leq T$. Since $T \asymp_B Y^2$, the Erdős–Turán inequality [3] gives the discrepancy of $\{D_y(N)\}$:

$$D_X^{\mathrm{sieve}} \ll_B T^{-1} + \sum_{\xi=1}^{T} \frac{|\mathbb{E}\mathrm{e}(\xi D_y(N))|}{\xi} \ll_B \frac{\log Y}{Y^2}.$$

Enlarging or shrinking each test interval by τ_X gives discrepancy $O_B(D_X^{\mathrm{sieve}} + \tau_X + Y^{-1}) = O_B(Y^{-1})$ for $\{D(N)\}$. Thus the exceptional fraction enters only once, after the Fourier estimate, rather than once per frequency.

Finally decompose a prefix up to M into dyadic blocks, discarding its first $O(\sqrt{M})$ indices. On the remaining blocks $\log\log X = \log\log M + O(1)$; endpoint rounding costs $O(\log M/M)$. Weighted averaging proves the stated ordinary rate. Applying this to base-B digit intervals proves normality. □

The constants and lower thresholds in these asymptotic estimates may depend on the fixed base B.

References

[1] Y.-G. Chen and I. Z. Ruzsa, *On the irrationality of certain series*, Period. Math. Hungar. **38** (1999), 31–37, doi:10.1023/A:1004742930674.

[2] T. F. Bloom, *Erdős Problem 259*, `https://www.erdosproblems.com/259`.

[3] L. Kuipers and H. Niederreiter, *Uniform Distribution of Sequences*, Wiley, 1974.